

Qixin Xu

Email ◇ Google Scholar ◇ Github

EDUCATION

Tsinghua University

2023 - 2027(expected)

B.S. in Computer Science; GPA: 3.9/4.0; Rank: 9/172

- **Selected A⁺/A Courses:** Linear Algebra, Probability and Statistics, Introduction to Artificial Intelligence, Introduction to Statistical Learning, Deep Learning, Multi-Modal Machine Learning, Programming and Training, Data Mining, Writing and Communication, English for Academic Purposes: Research Paper Writing etc.
- **Academic Interests:** Reinforcement Learning, LLM (Agents)

RESEARCH PUBLICATION

* *EQUAL CONTRIBUTIONS*, † *CORRESPONDING AUTHOR*

(1) **Qixin Xu**^{*}, Haozhe Wang^{*}, Che Liu, Fangzhen Lin, Wenhui Chen[†]. **DocThinker: Towards Unified thinking in Documents**, *Under peer review*, 2026.

(2) **Qixin Xu**, Bohan Lyu, Zihan Zhu, Yesai Wu, Zhong Zhang, Haotian Chen, Xin Cong, Xiaojiang Liu, Zhiyuan Liu[†], Maosong Sun. **Diversity-aware Training for Test-time Scaling**, *Under peer review*, 2026.

(3) Haozhe Wang^{*}, **Qixin Xu**^{*}, Changpeng Wang, Taofeng Xue, Chong Peng, Wenhui Chen[†], Fangzhen Lin[†]. **Bad Seeing or Bad Thinking? Rewarding Perception-Reasoning Synergy for Scalable Vision-Language Reasoning**, *ICML 2026 (Spotlight)*

(4) Ganqu Cui^{*}, Lifan Yuan^{*}, Zefan Wang^{*}, Hanbin Wang^{*}, Wendi Li^{*}, Bingxiang He^{*}, Yuchen Fan^{*}, **Qixin Xu**^{*}, Tianyu Yu^{*}, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, Ning Ding[†]. **Process reinforcement through implicit rewards**, *TMLR 2026*

(5) Haozhe Wang, **Qixin Xu**, Che Liu, Junhong Wu, Fangzhen Lin[†], Wenhui Chen[†]. **Emergent hierarchical reasoning in llms through reinforcement learning**, *ICLR 2026*

(6) Haozhe Wang, Haoran Que, **Qixin Xu**, Minghao Liu, Wangchunshu Zhou, Jiazhan Feng, Wanjuan Zhong, Wei Ye, Tong Yang, Wenhao Huang, Ge Zhang[†], Fangzhen Lin[†]. **Reverse-Engineered Reasoning for Open-Ended Generation**, *ICLR 2026*

(7) Tianyu Yu, Haoye Zhang, Qiming Li, **Qixin Xu**, Yuan Yao[†], Da Chen, Xiaoman Lu, Ganqu Cui, Yunkai Dang, Taiwen He, Xiaocheng Feng, Jun Song, Bo Zheng, Zhiyuan Liu[†], Tat-Seng Chua, Maosong Sun. **RLAIF-V: Open-Source AI Feedback Leads to Super GPT-4V Trustworthiness**, *CVPR 2025 (Highlight)*

(8) Pan Lu^{*}, Yifan Gao^{*}, William G Peng, Haoxiang Zhang, Kunlun Zhu, Elektra K Robinson, Qixin Xu, Masakazu Kotaka, Harrison G Zhang, Bingxuan Li, Anthony L Shiver, Yejin Choi, Kerwyn Casey Huang, Justin L Sonnenburg[†], James Zou[†]. **Eubiota: Modular Agentic AI for Autonomous Discovery in the Gut Microbiome**, *Submission to Nature*

RESEARCH EXPERIENCE

Stanford NLP Group

November 2025 – Present

Research Intern

- **Multi-Agent Systems for Scientific Discovery via Reinforcement Learning:** Developing a multi-agent Reinforcement Learning framework and infrastructure for autonomous scientific discovery; achieved State-of-the-Art performance on domain-specific benchmarks and validated findings through downstream human evaluation.

TIGER AI Lab

June 2025 – Present

Research Intern

- **Unlock the Implicit Mechanism of Reinforcement Learning:** Acknowledged two-stage training dynamics in RL, providing interpretability for black-box phenomena like “Aha moments” and “Length Scaling Effects”. Proposed Hierarchy-Aware Credit Assignment, outperforming GRPO across diverse model families. Paper was accepted by *ICLR 2026*.
- **Multimodal reasoning under stress tests and its mechanisms:** Developed coarse-to-fine algorithms and curated high-quality multimodal datasets, achieving state-of-the-art results on document reasoning benchmarks. Demystified the coupling between visual perception and text reasoning, improving general-purpose visual reasoning. Paper was accepted by *ICML 2026 (Spotlight)*.

Tsinghua NLP Lab

July 2024 – November 2025

Research Intern (advised by Prof. Zhiyuan Liu)

- **Enhancing Diversity in Large Language Models:** Identified that SFT and RL pipelines contribute to reduced diversity (mode collapse) in generation space; developed a diversity-aware data pipeline and integrated noisy embeddings as an implicit diversity signal, improving pass@k and other diversity metrics.
- **Enhancing Visual Reasoning in Multimodal Language Models:** Explored tool-use RL for visually intensive tasks (counting, grounding); developed a Tool-Use RL framework in veRL; improved Qwen2.5-VL-7B-Instruct on key counting/grounding benchmarks.
- **PRIME: Implicit Reward Modeling for Reasoning:** Proposed PRIME (**P**rocess **R**einforcement through **I**mplicit **r**Ewards) enabling online PRM updates using only policy rollouts and outcome labels via implicit process rewards; studied the role of dense rewards in RL. Paper was accepted by *TMLR 2026*.
- **Multi-modal Alignment through AI Feedback:** Investigated feedback from open-source models to reduce VLM hallucination; used a DPO-trained model as reward model for inference-time scaling. Paper was accepted by *CVPR 2025 (Highlight)*.

PROJECT EXPERIENCE

Contribute to the development of MiniCPM-V 2.6, MiniCPM-o 2.6 and MiniCPM-V 4.5 July 2024 - January 2025

Worked with OpenBMB Team

- Develop a simple yet effective pipeline to reduce MiniCPM-V 2.6 and MiniCPM-o 2.6’s hallucination
- Design particular format of pre-training data to enable the training of streaming ability of MiniCPM-o 2.6
- Technical report was accepted by *CVPR 2026*

SKILLS

Languages: English(CET-6: 651/710 (Listening: 249/249)), Chinese.

Programming: Python, C/C++, System Verilog, LATEX, etc.

Tools: vllm, Pytorch, DeepSpeed, Git, Shell, Vim, Django, ssh, gdb/pdb, etc.

Frameworks: veRL, Slime, EasyR1, LLaMA-Factory